Sumedh Kothari

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EDUCATION

Georgia Institute of Technology **Expected May 2028**
B.S. in Chemical and Biomolecular Engineering GPA: 3.91
Minors in Computer Science (Artificial Intelligence) and Physics (Quantum Sciences and Technology)

- **Coursework:** Computer Organization and Programming; Data Structures and Algorithms; Differential Equations
- **Honors:** PACESHIP Scholar, TechHack Valley 2026 Hackathon Winner, Symposium of Rising Scholars Presenter, Eagle Scout

EXPERIENCE

Witness AI Mountain View, CA
Machine Learning Intern May – September 2026

- Generated 10K jailbreak prompts (60% attack success on Claude, Qwen, Gemma) to train a prompt-safety classifier by building an adversarial self-play framework with SFT, generative adversarial training, GRPO-based RLVR, and Thompson sampling
- Converted 1,000+ arXiv papers into novel attack strategies and adversarial data for training and evaluating LLM safety classifiers and jailbreak guards by building LiLAC (Living Literature Adversarial Corpus), a fully autonomous agentic research pipeline
- Benchmarked 8 state-of-the-art open-source and proprietary LLMs and fine-tuned 3 safety classifiers on 3,000+ synthesized adversarial prompts by developing a scalable evaluation framework that also automated customer-facing LLM safety reports

ChemE Cube | Georgia Tech Atlanta, GA
Software, Mechanical, & Electrical Lead August 2025 – Present

- Accelerated CO_2-capture reactor optimization by developing a Gaussian-process Bayesian optimization framework that learns a surrogate model of capture performance from experimental data and selects the highest-value operating conditions to test next
- Delivered a fully fabricated, test-ready 1 ft^3 direct air capture (DAC) reactor for the AIChE ChemE Cube competition (Nov 2026) by leading a 10-member mechanical subteam through end-to-end CAD, prototyping, fabrication, and full systems integration
- Achieved a 75% reduction in ambient CO_2 concentration in bench-scale validation of the full direct air capture prototype by designing its airflow distribution and sorbent containment subsystems with parametric CAD, 3D printing, and CNC machining

RESEARCH

Ramprasad Laboratory | Georgia Tech Atlanta, GA
Machine Learning Researcher August 2026 – Present

- **PolyFrameNet: Block-Factored Polymer MLIP**
  - Built a polymer MLIP that enforces exact O(3) and permutation symmetry with plain MLPs, avoiding costly equivariant tensor products, by splitting chains into graph-defined rigid blocks with local frames and Weisfeiler–Lehman frame averaging
  - Achieved 2.5–5.5× faster inference and 3.4× less GPU memory than MACE and Allegro at an equal training budget (90 min, one L40S) on 876K-atom OPoly26; matched MACE force error on polyethylene (13.7 meV/Å) and scaled to 26K-atom melts
  - Open-sourced PyTorch/ASE code and weights with an 81-test correctness suite and a registry flagging unseen chemical blocks
- **PolyDYN** — Building a multi-agent LLM framework for autonomous polymer design with MLIP-driven property screening

PROJECTS

PolyT5-RLVR: Polymer Encoder-Decoder Model Summer 2026

- Reduced the rate of chemically invalid polymer generations 79% by reproducing the Ramprasad Lab's PolyT5 (7.5M-parameter T5 encoder-decoder pretrained on 92.3M polymers) and post-training it with GRPO on chemistry-verifiable rewards (RLVR)
- Cut Tg prediction error 5.2% (28.60 → 27.11 K) and structurally eliminated catastrophic decode failures by replacing text decoding with a linear regression head; the gain held for both random seeds on all 10 data splits of a 70-run, five-factor ablation
- Added split-conformal Tg prediction intervals (±54.35 K at 90% coverage) with a finite-sample marginal coverage guarantee

Machine Learning-Assisted Particle Swarm Optimization Spring 2026

- Awarded a $1,000 PACESHIP grant to research surrogate-assisted particle swarm optimization (PSO) for costly black-box tasks
- Beat the baseline PSO by 97.8% (Sphere), 99.9% (Rastrigin), 90.4% (Ackley), and 58.8% (Rosenbrock) with a hybrid optimizer using GP surrogates, LightGBM-guided repositioning of stagnating particles, and adaptive exploration–exploitation balancing
- Developed an autoresearch loop that autonomously ran experiments, tuned hyperparameters, and refined its search strategies

Panoptic Pigskin: NFL 4D Gaussian Splatting Spring 2026

- Reconstructed 3D NFL plays from just two broadcast angles at 0.15 m median error vs. NFL tracking data (41× below per-frame calibration) by self-training YOLOv8x-pose per play on 115K SMPL-X pseudo-labels projected into both pose-solved cameras
- Cut physically implausible motion from 17% to 5% of frames by smoothing each track along its camera's depth axis, powering free-viewpoint Gaussian-splat replays and automated grading of all 22 players (separation, time to throw, air yards, and YAC)

SKILLS

- **Programming:** Python, C++, CUDA, Java, MATLAB, Linux, Bash
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace Transformers, vLLM, scikit-learn, LightGBM, OpenCV
- **Atomistic Simulation:** ASE, LAMMPS, VASP, MACE, Allegro, UMA
- **CAD & Fabrication:** PCB Design (KiCAD), SolidWorks, Fusion, 3D printing, CNC machining, Soldering
- **Languages:** English, Hindi, Spanish